pysie2d

Quasi-normal mode extraction

Methods and validation strategy

The implemented state of the feature: the contour-integral eigensolver, the analytic anchor it is validated against, and the evidence behind each claim. Every number quoted here was measured against the code, not recalled.

Implemented

- **Phase 1** Analytic Mie QNM roots + independent completeness check
- **Phase 2** Beyn contour eigensolver, on pencils of known spectrum
- **Phase 3** QNMSolver facade, validated against the analytic anchor

Not implemented

- ○ **Phase 0** vacuum-wavelength convention fix (ships separately, v0.4.0)
- ○ **Phase 4** analytic dM/dλ
- ○ **Phase 5** QNMResult.refine(), σ_min by inverse iteration

Status: the Phase-3 go/no-go passed - both polarisations reproduce the analytic poles at first order, degeneracy counted correctly. Two limits found in review since (spurious resonances, a Q ceiling) are recorded in sections 8 and 9.

26 July 2026

Suite: 73 tests, 36 s. Source: docs/design/beyn-port-spec.md, docs/conventions.md §8.

1 What the feature computes

A quasi-normal mode is a source-free solution of the scattering problem: a complex wavelength at which the boundary-integral operator M(λ) is singular, so a field can persist on the particle with no incident wave driving it. Because the mode radiates, it decays in time, and the wavelength at which it sits is genuinely complex. The real part locates the resonance; the imaginary part sets how long it lives.

The quantity users want from it is the quality factor, which the result object exposes as a derived property:

$$Q \;=\; \mathrm{Re}\,\lambda \,/\, (2\,\mathrm{Im}\,\lambda) \;=\; -\;\mathrm{Re}\,\omega \,/\, (2\,\mathrm{Im}\,\omega)$$

Finding these modes means finding where a matrix-valued function of a complex variable becomes singular. That is a nonlinear eigenvalue problem: λ enters M through Hankel functions, not linearly, so there is no matrix whose eigenvalues are the answer. The classical approach - scan a grid, look for dips in the smallest singular value, polish the dips - needs an initial guess per mode, silently misses modes between grid points, and degrades exactly where the physics is most interesting, near high-Q modes whose dips are narrowest.

Why contour integration instead

Beyn's method replaces the search with an integral. If M is holomorphic on and inside a closed contour, then integrating $M(\lambda)^{-1}$ against a random probe matrix V picks out precisely the modes enclosed:

$$A_0 = \frac{1}{2\pi i}\oint M(\lambda)^{-1} V\, d\lambda, \qquad A_1 = \frac{1}{2\pi i}\oint \lambda\, M(\lambda)^{-1} V\, d\lambda$$

The rank of A_0 equals the number of modes inside the contour, counted with multiplicity, provided those modes are semisimple - which for a circle they are, the ±n degeneracy being symmetry-protected. The pair (A_0, A_1) then reduces to a small ordinary eigenvalue problem whose spectrum is exactly that set. No initial guess, no scan, no bracketing: the contour decides what is found, and the cost is fixed at one assembly and one factorisation per quadrature node regardless of how many modes are inside.

> The price of that guarantee is holomorphy. Everything in the conventions section below exists to keep M(λ) holomorphic on the search rectangle - it is the premise of the method, not a detail.

2 Conventions the implementation pins

Every one of these is recorded in docs/conventions.md §8 and asserted somewhere in the suite, because each is a place where a plausible-looking sign or factor produces a plausible-looking wrong answer.

The decaying half-plane

Under the package's exp(-iωt) time convention with outgoing $H^{(1)}$ waves, a decaying mode has Im ω < 0, hence Im k < 0, hence:

$$\mathrm{Im}\,\lambda \;>\; 0$$

A search box must therefore lie strictly in the upper half λ-plane, and strictly in Re λ > 0. Both bounds raise ValueError rather than being left to the user's care.

> The Re $\lambda > 0$ bound is sufficient but not necessary, and the reason first written down for it was wrong. In a homogeneous background $M(\lambda)$ is holomorphic on the whole cut plane - $H^{(1)}$ is analytic for arg λ in $(-\pi, \pi)$, not $(-\pi/2, \pi/2)$ - so a box straying to negative Re λ does not break holomorphy. The real reason to forbid it is that it is the mirror region, so it duplicates physics, and that it straddles the branch point at the origin. The guard is right; its stated justification is being corrected.

> The research code this was ported from carries the opposite sign. It is wrong relative to the operator it calls, and was deliberately not ported - one of the few places where the port rejects its source.

Poles do not come in conjugate pairs

The reality condition for a real, non-dispersive scatterer is $\omega \to -\bar{\omega}$, that is $\lambda \to -\bar{\lambda}$, not $\lambda \to \bar{\lambda}$. Expecting conjugate pairs is the natural mistake for anyone carrying real-eigenvalue intuition into a non-Hermitian problem, so the suite carries an explicit negative test: at the conjugate of a genuine mode the operator is no more singular than at an arbitrary point (measured 7.2×10^{-3} against 4.3×10^{-4} at the mode).

> The mirror partner at $-\bar{\lambda}$ is the negative-frequency image of the same physical mode: it carries no independent information, and it is not a singularity of this operator either. Measured σ_min/σ_max at $-\bar{\lambda}$ is 4.7×10^{-3}, more regular than a generic point in the search box. $\bar{\lambda}$ is a mode only of the conjugate, $\exp(+i\omega t)$ equation. Saying that mirror partners 'sit at negative Re λ' invites a reader to look for them there and find nothing.

Degeneracy is structural

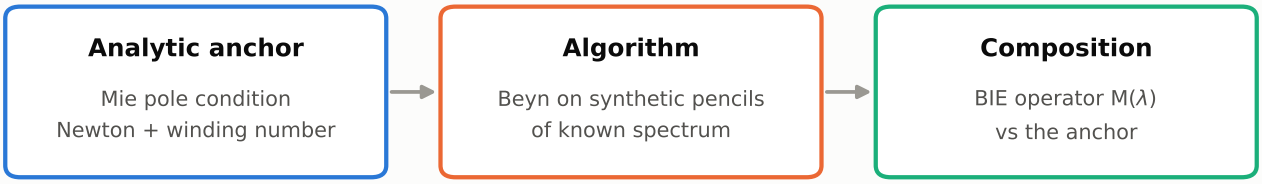
Every azimuthal order $n \geq 1$ of a circle is doubly degenerate through $\exp(\pm in\theta)$; only $n = 0$ is simple. Degenerate partners are reported as two separate entries carrying multiplicity 2, never collapsed into one - the pair spans two independent mode vectors, and merging them would make the mode count disagree with the analytic one.

Polarisation

pol = 2 is TE, mapping to the Mie coefficient b_n; pol = 1 is TM, mapping to a_n. This is the package's existing convention and the QNM code inherits it unchanged; the suite anchors both polarisations separately so a swapped code cannot pass.

3 The validation strategy

The package's standing rule is that new physics needs an independent anchor - a closed form, an analytic limit, or a second method - and explicitly not agreement between two paths in the same repository, which only proves they share assumptions. The strategy here is a chain of three links, each checked by a method that does not share the failure mode of the link beside it.



each link is checked by a method that does not share the failure mode of the link beside it

The separation is what makes a failure diagnosable. On a circle, a wrong pole could mean a broken contour integral, a mis-detected rank, a sign convention, or a discretisation error, and those are hard to tell apart from one number. Validating the algorithm on pencils of known spectrum first - where there is no electromagnetics to get wrong - means a later disagreement on the circle can only be physics or convention.

> Concretely: Phase 2's tests never touch Maxwell, and Phase 1's tests never touch the BIE solver. Only Phase 3 composes them, and by then each half has its own independent evidence.

4 Link one: the analytic anchor

For a circular cylinder the mode condition is available in closed form. The Mie coefficients a_n and b_n are ratios, and a pole of the coefficient is a zero of its denominator - a source-free solution. Taken verbatim from the coefficient definitions in the same module, with x the size parameter and m the relative index:

$$D_n^{TM}(x) = m J_n(mx) H_n'(x) - J_n'(mx) H_n(x)$$

$$D_n^{TE}(x) = J_n(mx) H_n'(x) - m J_n'(mx) H_n(x)$$

These depend only on (x, m), so the modes are radius-independent; the radius only maps them onto a wavelength through $\lambda = 2\pi a/x$. Roots are found by seeding from a coarse complex grid and polishing with a secant Newton iteration.

Two design decisions worth naming

- The seeding runs on a scale-free residual, D divided by the magnitude of the two terms that cancel at a root, rather than on |D| itself. The raw denominator spans many orders of magnitude across a search window - it inherits the exponential growth of H_n at large order and small argument - so |D| has no usable grid minima at all. Normalising turns a root into a clean minimum of a quantity of order one, and makes a single residual tolerance meaningful at every order.
- The grid's Im-x floor is an argument, not a constant, because it is a ceiling on Q: a mode with Q above Re x / (2 · floor) sits below the lowest row sampled and cannot be seen. In this spectrum the TE ladder runs Q = 164, 598, 2289 for n = 4, 5, 6 - Im x falling about 3.4× per order - so two more orders would have crossed a fixed floor silently.

Why the root list is believed to be complete

A grid-seeded search can only miss roots, and its own report of completeness is worth nothing. The count is therefore checked against the winding number of D_n along the box boundary, which counts zeros with multiplicity by the argument principle and is computed without reusing the root-finder at all. It agrees for every order n = 0 to 14 in both polarisations - and it is what revealed a third TM n = 2 root, a very broad Q = 0.96 mode that the original design document had missed.

That extra root is not a third radial branch, as the design document first called it. It belongs to a broad, barely-confined family that exists at every order in both polarisations, whose |Im x| grows with n so that only its lowest members rise above the search box's floor - TM at n = 2, 3, 4 sits at |Im x| = 0.91, 0.99, 2.55. The anchor table's very first row, TM n = 3 at Q = 1.41, is a second member of the same family. The consequence for testing is the one that matters: the root count per order is set by where the box cuts a smooth family, not by the order, so a count can only ever be asserted from the measured table.

A second, cheaper check guards the polarisation mapping: the identity $D^{TM}_0 \equiv D^{TE}_1$ holds algebraically (measured to 1.4×10^{-15} over 200 random complex points), and no numerical search can fake it. A swapped pol code breaks it immediately.

Analytic QNM spectrum of the anchor circle, and the three search boxes

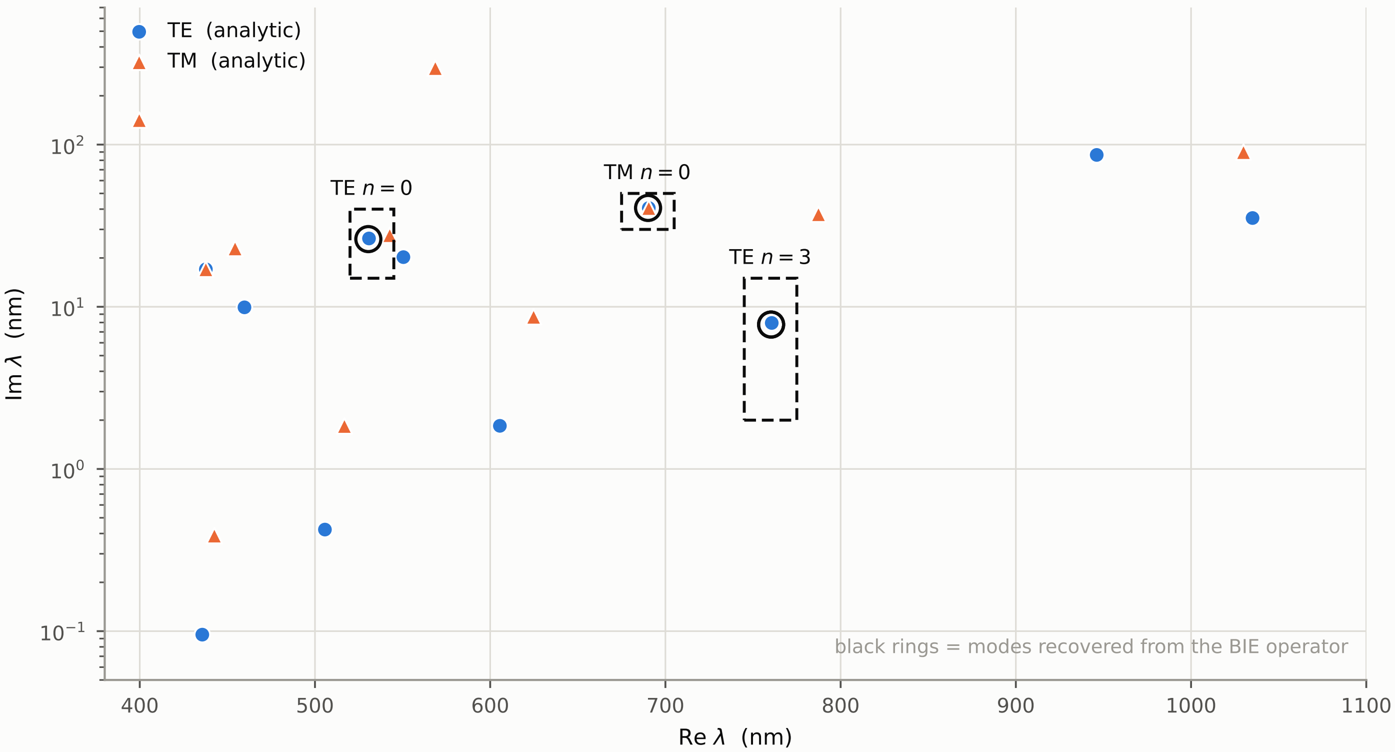


Figure 1. The 22 analytic modes of the anchor circle (radius 200 nm, n_core = 3.0, in vacuum) in the complex wavelength plane, on a logarithmic Im axis spanning Q = 0.96 to Q = 2289. Dashed rectangles are the three search boxes used for validation; black rings mark the modes recovered independently from the BIE operator. The TE n=0 box is deliberately tight in Im: two other TE modes sit within 25 nm in Re, and only Im separates them.

5 Link two: the algorithm, on known spectra

The eigensolver is deliberately free of electromagnetics. It sees a callable $\lambda \to M(\lambda)$ and a rectangle, which is what makes it testable against pencils built from a diagonal of known roots conjugated by unitary factors - there, nothing can be wrong except the algorithm. Both a linear pencil and a genuinely nonlinear one are used, the latter because a linear pencil would pass even if the method had silently degenerated into an ordinary matrix eigenvalue solve.

Quadrature

A rectangular contour with Gauss-Legendre nodes per edge. Gauss-Legendre rather than the trapezoid because the integrand is analytic on each straight edge but not periodic along it, so the trapezoid rule's spectral accuracy does not apply while Gauss-Legendre's does. The $1/(2\pi i)$ and the parametrisation Jacobian are folded into the returned weights, so the suite's Cauchy test exercises the normalisation itself rather than its caller.

Rank detection: the last cliff, not the largest

Exactly, A_0 has rank equal to the mode count; quadrature error lifts the remaining singular values into a noise floor. The rank is read off the last gap that clears a factor of 10^3 and whose retained value is still above rank_tol - not the largest gap. Two poles whose residues differ by 10^6 produce two cliffs, and the first can be the taller: measured gaps of 1.1×10^6 then 2.1×10^5, where taking the maximum returns rank 1 and loses the weak mode entirely. Everything past the final cliff is flat noise, which is what a floor is; a drop between two genuine poles is not.

Telling an empty box from a saturated probe

Both produce the same flat singular spectrum, so the natural fallback - report no modes - is a confident wrong answer when the probe is too narrow. The discriminator is absolute rather than relative: over an empty contour the integrand is analytic and A_0 cancels to nothing, while a saturated probe returns a perfectly large A_0. The solver therefore also reports

$$\text{cancellation} = \|A_0\| \,/\, \sum_j |w_j|\,\|x_j\| \qquad (\leq 1)$$

and refuses, loudly, when the spectrum is flat but A_0 did not cancel. The two sides scale differently, which is what makes a threshold possible at all: empty-box cancellation is quadrature error and falls with the node count, while a residue enclosed by the contour does not move.

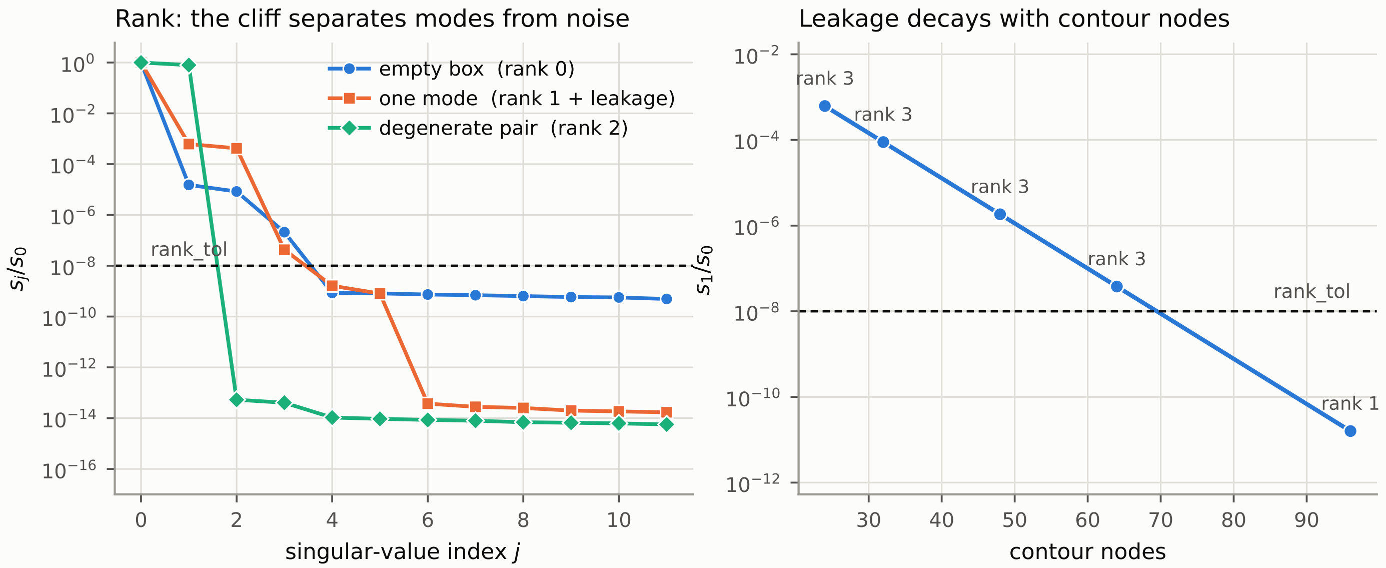


Figure 2. Left: singular-value spectra for the three regimes, from the real BIE operator. The rank is the index of the last cliff; below rank_tol the values are quadrature noise. Right: a pole 5 nm outside the contour leaks rank into it, and that leakage decays geometrically with quadrature resolution - by 96 nodes it falls below rank_tol and the reported rank drops to its correct value of 1.

6 Link three: composition, and the go/no-go

The facade constructs the same way as the driven solver and calls BIESolver.assemble itself, on its contour, rather than duplicating the assembly. That is deliberate: the claim being made is that a QNM is a singularity of the scattering operator, and it only holds if the two are literally the same matrix.

Three anchors were chosen to exercise different failure modes: a simple mode crowded in Re λ, a degenerate pair, and a mode in the other polarisation.

anchor	analytic λ (nm)	recovered λ (nm)	ΔRe	ΔIm	K	rank
TE n=0	530.832+26.378j	530.456+26.138j	-0.377	-0.240	1	3
TE n=3	760.687+7.948j	760.326+7.770j	-0.361	-0.178	2	2
TM n=0	690.514+40.642j	690.105+40.682j	-0.409	+0.040	1	1

All at 200 boundary points. The errors are discretisation error in the BIE solver, not extraction error - which is the point of measuring the convergence order rather than resting on a tolerance.

Pole error is first order in resolution

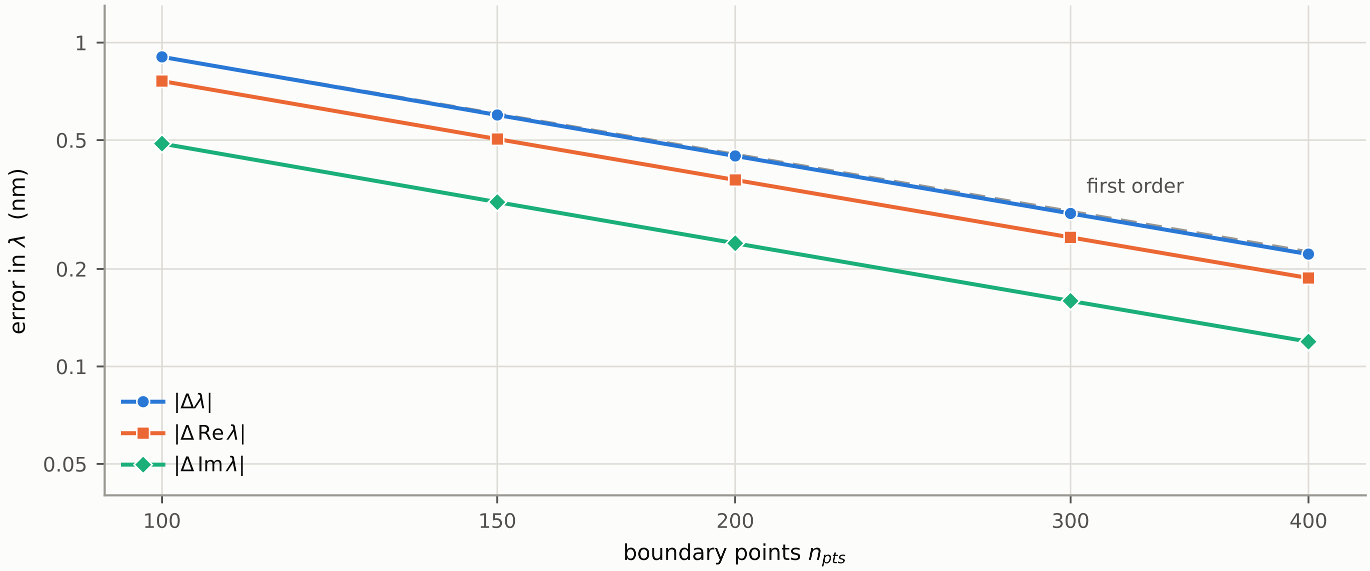


Figure 3. Convergence of the recovered TE n=0 mode against the analytic value, with the first-order reference slope. Both parts of λ converge at order 1.00-1.03; the fitted tolerances in the suite are an error budget derived from this, sitting about 30% above what 200 points delivers.

Measuring the order is what makes those tolerances falsifiable. A pole that stagnated, or converged to something other than the anchor, would still pass a fixed tolerance if the tolerance were loose enough; asserting the rate excludes both.

Two stronger checks, added after review

Matching three anchors one at a time shows the extraction finds modes that are there. It does not show it fails to invent modes that are not - the classic failure of a second-kind boundary formulation is a spurious interior resonance, and a single-mode box cannot see one. A wide box does. Over Re λ from 600 to 800 nm and Im λ from 1 to 60 nm - a region whose size-parameter image lies strictly inside the analytically complete box, so the table is known to be exhaustive there - the analytic spectrum holds three distinct TE modes, each doubly degenerate, so six with multiplicity:

```
analytic:  605.52336+1.84489j   690.51371+40.64175j   760.68665+7.94771j
recovered: 604.92376+1.54535j   690.38584+40.62731j   760.08703+7.65097j
K = 6, rank = 6      sv: 1.0  0.84  0.67  0.31  0.27  0.16  1.7e-14 ...
                     (n_pts = 120, 64 contour nodes, n_probe = 14)
```

Six modes, rank six, nothing spurious, and the singular spectrum falls off a cliff of thirteen decades exactly at index six. That is the check that rules out spurious interior resonances, and it costs about twelve seconds.

The second check turns the $D^{TM}{}_0 \equiv D^{TE}{}_1$ identity into a statement about the BIE operator rather than about the analytic formula. The identity says TM n=0 and TE n=1 sit at the same wavelength - but TM n=0 is simple and TE n=1 is degenerate, so one box searched in two polarisations must return different counts:

```
box 675+30j -> 705+50j, n_pts = 120
  pol = 1 (TM):  K = 1,  rank = 1,  multiplicity [1]     689.828+40.708j
  pol = 2 (TE):  K = 2,  rank = 2,  multiplicity [2,2]   690.386+40.627j
```

One wavelength, two polarisations, multiplicity 1 against 2. A swapped polarisation code in the assembly would break this, and nothing else in the suite tests that at the mode level.

Diagnostics the result carries

Each field distinguishes a named failure mode rather than describing the run:

- sv_ratio, max_gap and rank - whether the box has meromorphic content at all. rank may legitimately exceed the mode count; see below.
- edge_margin, as a fraction of the shorter side - a mode close to a contour edge is being clipped and its value is not trustworthy. The shipped boxes report 42%, 27% and 33%.
- sigma_ratio = σ_min/σ_max at each mode, never an absolute σ_min, which carries the scale of the operator and is therefore dimensionally meaningless as a gate. Measured: 8×10^{-9} at a recovered mode against $2\text{-}5\times10^{-3}$ at generic points.
- cancellation - empty box versus saturated probe, as above.

> sigma_ratio is a ruler, not a threshold detector - see below. Reading it as a pass/fail gate was the mistake that put a wrong sentence about it in the design document.

What sigma_ratio actually measures

Measured along a ray from the discrete pole, σ_min/σ_max is linear in the distance to it, at 9.5×10^{-4} per nm, constant to 0.5% over three decades of $|\Delta\lambda|$ and saturating near 4×10^{-3} beyond about 5 nm:

|Δλ| (nm)	**0.0045**	**0.0134**	**0.045**	**0.134**	**0.447**	**1.34**
σ_min/σ_max	4.3e-6	1.3e-5	4.3e-5	1.3e-4	4.3e-4	1.3e-3
ratio / |Δλ|	9.54e-4	9.54e-4	9.54e-4	9.55e-4	9.58e-4	9.67e-4

The analytic pole reads 4.3×10^{-4} purely because it sits 0.447 nm from the discrete one, and the linear law returns that distance exactly. So it is not 'no better than a generic point' - it is 5.6× more singular than the nearest generic sample, and reads precisely what its distance dictates. The gap is the discretisation error quantified rather than merely visible, and it gives the 10^{-6} threshold a concrete meaning: within about 10^{-3} nm of the discrete pole.

7 What implementation changed about the plan

The design document was written before the code and measured against a lower-contrast test case. Seven of its claims did not survive contact with the implementation. They are listed here because the corrections are the substantive part of the strategy, not errata.

Reduced pencil
: Σ_1^{-1} post-multiplies: $B = U_1^H A_1 V_1 \Sigma_1^{-1}$. The drafted left-multiplication gives a similar matrix, so eigenvalues survive the error and eigenvectors do not - a test on λ alone would never have caught it.

Rank rule
: Last qualifying gap, not the largest. The drafted argmax rule fails on disparate residues, and also fails the document's own degenerate example.

Saturation check
: The drafted check could never fire - rank is at most p-1 by construction. Replaced by the cancellation diagnostic.

Cancellation floor
: Calibrated at 10^{-10} on synthetic pencils, which cancel to 10^{-16}. Physical contours reach 5×10^{-7}, and that produced a spurious saturation error on a genuinely empty box. Now 10^{-4}.

Clean rank gap

Predicted on the grounds that pysie2d assembles exactly. The cause is quadrature error, not assembly. Rank may exceed the mode count by two distinct mechanisms: leakage from a pole just outside the contour, and ordinary quadrature error on the analytic part of $M(\lambda)^{-1}V$, which does not vanish either - rank 4 was measured for 2 modes in a box with no other pole within 150 nm.

Anisotropic error

Claimed Q converges far better than Re λ, measured at m = 1.5. At m = 3.0 both parts are first order with a constant ratio, and since Im λ is small its relative error is the worse of the two - Q converges worse, not better: 0.87% at 200 points.

Cauchy tolerance

Holds at 10^{-13} for 96 nodes, not the default 48, and depends on how far the pole sits from the contour.

8 Limits of what is implemented

- Homogeneous background only. A slab or layered background breaks the holomorphy argument that licenses the contour method, which is one reason it stays out of scope rather than being an unfinished feature.
- Mode vectors are exposed raw, in the boundary-field layout of the driven solver. They are not normalised as mode fields - that needs a QNM norm, which is not implemented.
- Degenerate modes are found and counted correctly, but cannot be refined: the bordered-Newton system assumes a simple eigenvalue and its Jacobian is singular for a two-dimensional null space. The condition number is reported rather than the failure being silent.
- Refinement is not exposed at all yet. The bordered-Newton core exists and is tested on synthetic pencils, but it needs an analytic dM/dλ, which is Phase 4.
- Modes are those of a non-dispersive dielectric. This is a precondition of the physics, not only of the derivative Phase 4 will add: with n(ω) the mode condition is a different nonlinear problem and these are not its solutions.
- Defective eigenvalues - exceptional points, reachable on a non-circular boundary - are outside the guarantee. The rank argument holds for semisimple modes; at a Jordan block the recovered wavelength degrades from order ε to order √ε, and nothing in the suite currently exercises that case.
- Not every singularity found is a quasi-normal mode. Interior Dirichlet resonances of the formulation sit near Im λ = 0 with excellent diagnostics, and no field of the result distinguishes them from physical modes - see section 9.
- Quality factors above roughly n_pts/10 are not resolved. Q inherits the error of Im λ, which is small precisely when Q is large - section 9.
- Public wavelengths are documented as vacuum but implemented as medium wavelengths. The two coincide only at n_clad = 1, which is where every test - including this anchor - runs. Reconciling them is a breaking change to the driven solver, shipping separately.

> The last item is a live trap rather than an inconsistency on paper. QNMSolver documents vacuum wavelengths and carries no guard, so at n_clad ≠ 1 it returns wavelengths wrong by that factor with no symptom - and the analytic reference returns medium wavelengths, so the two disagree by n_clad as well. The validation deliberately runs at the one background index where this is invisible. Refusing n_clad ≠ 1 outright is the recommended stopgap until the convention fix ships.

9 Independent review, and two limits it found

Two independent reviews were run over the implemented state: one on the electromagnetic theory and its numerical realisation, one on the modelling method and validation logic. Every claim reproduced below was re-measured against the code before being recorded here.

Confirmed

- The sign convention, verified against the operator rather than argued: the analytic pole reads $\sigma_{min}/\sigma_{max} = 4.3\times10^{-4}$, its conjugate 7.2×10^{-3}, a generic point 2.4×10^{-3}. The research code's opposite sign is inconsistent with the very kernels it calls, so rejecting it was right.
- The mode conditions and the polarisation map, re-derived from the continuity conditions rather than copied. Also that every zero of the denominator is a genuine pole - numerator and denominator share no zero, because their resultant is the Wronskian $2i/\pi x$, which never vanishes. That is the one-line justification the code asserted without proof.
- First order in n_pts is the expected rate, and now for a stated reason: the smooth part of the Nyström discretisation would be spectral, but the analytic log-singularity diagonal is $O(h)$, and a simple eigenvalue of a holomorphic family inherits the operator's error at first order.
- The Beyn reduction, the rank rule, the quality-factor ladders as whispering-gallery behaviour (Q growing geometrically in n once the mode is below the light line), and the leakage explanation - the leaked eigenvalues are the discrete TE n=2 pole to 10^{-10} nm, which is far stronger evidence than the 0.07 nm first recorded.

A spurious resonance the search region admits

The most consequential finding. Searching Re λ in [518, 527], Im λ in [0.1, 1.5] - a perfectly legal box - returns one mode with flawless diagnostics: rank 1, one mode, edge_margin clear, and $\sigma_{min}/\sigma_{max}$ of 6×10^{-16}, better than any genuine mode in this document. It is not in the analytic table, and the winding number says that table is complete. It is not a QNM:

```
n_pts = 100    522.4047 + 0.3989j     sigma_ratio 3.5e-16
n_pts = 200    522.4750 + 0.1960j     sigma_ratio 1.5e-16
n_pts = 300    522.4990 + 0.1299j     sigma_ratio 6.3e-16

Richardson ->  522.5471 - 0.0023j
2*pi*R/j_01 =  522.5481          (j_01 = 2.404826)

Mie denominators there:  |D_0| = 0.163   |D_1| = 0.462   |D_2| = 0.225
```

Im λ drives to zero as the mesh refines and the real part converges on $2\pi R/j_{0,1}$ to five figures. This is the interior Dirichlet resonance of the background-filled disc - an artefact of the integral formulation, not a physical mode. The Mie denominators there are of order one, so it is definitively not a pole of the analytic solution.

> The three anchors in this document are genuine Mie poles, verified against an independent analytic table, and none of the measured results change. What changes is the claim that can be made about the extractor: it finds every QNM in a box, and it may also find singularities of the formulation that are not QNMs. Nothing in QNMResult currently distinguishes them, and these sit at Im λ ≈ 0, interleaved with exactly the high-Q modes a user would be hunting.

A quality-factor ceiling

The relative error of Im λ, and therefore of Q, scales as C/(n_pts · Im λ). Since Im λ falls as Q rises, accuracy in Q degrades linearly with Q at fixed resolution. Measured on the TE n=5 mode of this same fixture, analytic Q = 597.7:

```
n_pts = 100     Q = 2945      +393 %
n_pts = 200     Q =  988.6     +65 %
n_pts = 400     Q =  744.2     +25 %
```

Holding Q to 10% needs roughly n_pts > 10·Q, so the fixture's Q = 2289 mode would need of order 10^5 boundary points - a dense matrix beyond reach. The fixture is advertised as spanning Q = 0.96 to 2289, and the extractor tests exercise none of it above Q = 48. The high-Q half of that range is unreachable in principle, and belongs in the API documentation as a formula rather than being discovered by a user.

Defects in the test suite itself

Three findings concern the evidence rather than the physics, and all three weaken claims this document makes:

- Five tolerances are 5-6 orders looser than written, because np.allclose carries rtol = 1e-5 by default and none override it. The seed-independence check states atol = 1e-9 nm and actually enforces 5×10^{-3} nm. The measured seed scatter is 1.5×10^{-6} nm, so the honest bound is rtol = 0 with atol = 1e-5.
- One assertion is a tautology. sv_ratio is normalised by its own largest entry, so sv_ratio[0] is exactly 1 and the 'independent' second check on leakage is character-for-character the first one.
- The LU-failure guard cannot fire. scipy's lu_factor does not raise on an exactly singular matrix - it warns, returns, and yields NaN, which propagates into the moments and finally surfaces as an unrelated SVD error. The documented 'skip the node and warn' path is unreachable; it needs an explicit finiteness check.

> None of these invalidate the three anchor results, which are reproduced above from the shipped code. They do mean the suite currently proves less than it appears to, and that is worth more than a passing test count.